Vapor Stone
RAIL SYSTEMS
A Wabtec company

10 655 Henri-Bourassa Blvd., We
Saint-Laurent (Québec) Canad
H4S 1A1

Tel: 514.335.4200
Fax: 514.335.4231
www.wabtec.com

Vapor Stone Rail Systems

Generic Software Documentation Project

Project # 7260

Door System Boot Loader software

Software Requirements Specifications (SRS)

Contract Number		CDRL Number	
		Customer Ref. Number	

NOTICE
Plans, specifications and engineering data submitted herewith or which may be submitted, remain the property of **Vapor Stone Rail Systems** and shall not be reproduced or altered or used in whole or in part except by express consent of **VSRS**.

AVIS
Les plans, spécifications et données techniques soumis ci-joints, ou qui peuvent être soumis, restent la propriété de **Vapor Stone Rail Systems** et ne doivent pas être reproduits, altérés ou utilisés, totalement ou partiellement, sans le consentement express de **VSRS**.

	Name / Title	Signature / Date
Prepared :	Thomas Marmeth Software Developer	08/05/2011
Verified :	Eric Malo, Eng. System engineer	2011-08-05
Approved :	Martin Ouellet, Eng. Software department supervisor	2011-08-05

Document No.	**201111271**	Revision	**1**
		Date (yyyy-mm-dd)	**2011-07-21**

Revision List

Rev.	Date	By	ECD	Description of changes
1	2011-07-21	T.Marmeth	---	Official release
0.1	2011-07-13	T.Marmeth	---	Implemented post-inspection comments from inspection n°I-1696-EMALO Suppressed “Send SCI versions” function, redundant with “Send software versions” function Suppressed “3.2.1.5 Receive file from PTA” and “3.2.1.6 Transfer file to PTA”, not used by Boot Loader.
0	2011-06-22	T.Marmeth	---	Original issue

Preface

This Software Requirements Specification (SRS) complies with the requirements of the IEEE 830-1998, IEEE Recommended Practice for Software Requirements Specifications, approved on 1998-06-25 and of the IEEE Std 1558-2004, IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems approved on 2005-05-23.

Vapor Stone Rail Systems
10655, Henri-Bourassa West
St-Laurent, QC, CANADA, H4S 1A1
ph. 514-335-4200
http://www.vaporrail.com/

Table of Contents

REVISION LIST **2**

PREFACE **3**

TABLE OF CONTENTS **4**

LIST OF FIGURES **6**

LIST OF TABLES **6**

1 INTRODUCTION **7**

1.1 Purpose 7
1.2 Scope 7
1.3 Definitions, Acronyms, and Abbreviations 7
1.3.1 Definitions 7
1.3.2 Acronyms 8
1.4 References 8
1.5 Overview 8

2 OVERALL DESCRIPTION **10**

2.1 Product Perspective 10
2.1.1 System Interfaces 10
2.1.2 User Interfaces 10
2.1.3 Hardware Interfaces 10
2.1.4 Software interfaces 11
2.1.5 Communications interfaces 11
2.1.6 Memory constraints 11
2.1.7 Operations 11
2.1.8 Site adaptation requirements 11
2.2 Product functions 12
2.3 User characteristics 12
2.4 Constraints 12
2.5 Assumptions and dependencies 12
2.6 Apportioning of requirements 12

3 SPECIFIC REQUIREMENTS **13**

3.1 External interfaces 13
3.1.1 HW_COM 13
3.1.2 Memory 15
3.1.3 LED 17
3.1.4 Inputs 17
3.2 Functions 18
3.2.1 PTA Communication Functions 18
3.2.2 Self-tests functions 22
3.2.3 User interface functions 23
3.3 Performance requirements 24
3.3.1 Static numerical requirements 24
3.3.2 Dynamic numerical requirements 24
3.4 Logical database requirements 24
3.5 Design constraints 24
3.5.1 Standards compliance 24
3.6 Software system attributes 24
3.6.1 Reliability 24
3.6.2 Availability 24
3.6.3 Security 24
3.6.4 Maintainability 24

3.6.5 Portability 25
3.7 Other requirements 25

List of figures

Figure 2-1: System interface diagram 10
Figure 3-1: Software product perspective 13

List of tables

Table 3-1: Mnemonics of HW_COM elements 14
Table 3-2: Mnemonics of memory elements 15
Table 3-3: MemoryApplication.DMSApplication.AppInfo and MemoryApplication.BootLoaderApplication.AppInfo data format 16
Table 3-4: Mnemonics of LED elements 17
Table 3-5: Mnemonic of inputs elements 17

1 INTRODUCTION

This document is the software requirements specifications for the Door System Boot Loader application developed by Vapor Stone Rail Systems.

This document lists the specifications required to verify the integrity and initialize the processor board of the Door Module System. The VAP2 Door Module System Boot Loader application is part of the microprocessor based system responsible for monitoring and controlling each door.

1.1 Purpose

The purpose of this Software Requirements Specifications document is to specify the requirements for implementing the Door Module System Boot Loader application that is part of the Door Module System.

The intended audience is the Vapor Stone Rail Systems software development and maintenance team members, employees of transit authorities and car builders using this software.

1.2 Scope

The Door Module System Boot Loader is a software application that resides inside the Door Module System. It is the startup program of the door module system. In case of no Valid Door Module System application software inside the application memory of the Door Module System or a user request by the Portable Test Application, the Door Module System Boot Loader initializes the Door Module System to be able to receive a new version of the application. The Door Module System Boot Loader interfaces with the Portable Test Application. The Door Module System Boot Loader does not enable door operation.

1.3 Definitions, Acronyms, and Abbreviations

Definitions and acronyms listed below establish meaning in the context of this SRS.

1.3.1 Definitions

- **Door Module System** — A processor based electronic box running software that controls the door, provides monitoring functions, and interacts with the user via a PTA and a network
- **Door Module System Boot Loader application:** — Software that verifies the hardware of the door module system, activates the warning and fault light emitting diodes (LED), reprograms application, communicates with the portable test application and starts the door module system application automatically.

1.3.2 Acronyms

CPU	Central Processing Unit
DMS	Door Module System
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers
I/O	Input / Output
LED	Light Emitting Diode
ms	millisecond
PTA	Portable Test Application
SCI	Software Configuration Item
SRS	Software Requirements Specifications
Std	Standard
TS	Technical Specification

1.4 References

This section lists the number, title, revision, date, and source of all documents referenced in this SRS.

Standards Documents

IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications, 1998-06-25, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, 2005-03-23, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

CMM-01	Software Engineering Process, VSRS, Rev. 3, 2001-03-31
CMMI-02	Software engineering policy manual, VSRS, Rev. 1, 2008-08-17
S17-01	Software Requirements Specification Template, Rev. 0, 2002-01-15
201111489	Portable Test Application -- Interface Design Description (IDD), Latest revision

1.5 Overview

The Software Requirement Specifications (SRS) document describes all inputs and outputs from the software system. It describes the functionalities and the relationship of these I/Os. It further identifies the responses to abnormal situations when processing these inputs and outputs. The SRS specifies the performance requirements and constraints imposed on the design of the Door Module System Door System Boot Loader software.

- Section 1: *Introduction,* gives an introduction, definitions, and overview of the SRS.
- Section 2: *Overall Description,* describes the factors that affect the product and its requirements.

- Section 3: *Specific requirements,* describes all the inputs/outputs relationships and functionalities.

2 OVERALL DESCRIPTION

2.1 Product Perspective

2.1.1 System Interfaces

The Door Module System (DMS) Boot Loader interfaces with the Portable Test Application (PTA). The interface with the PTA is done via a hardware communication interface (HW_COM) that can be USB, serial or Ethernet.

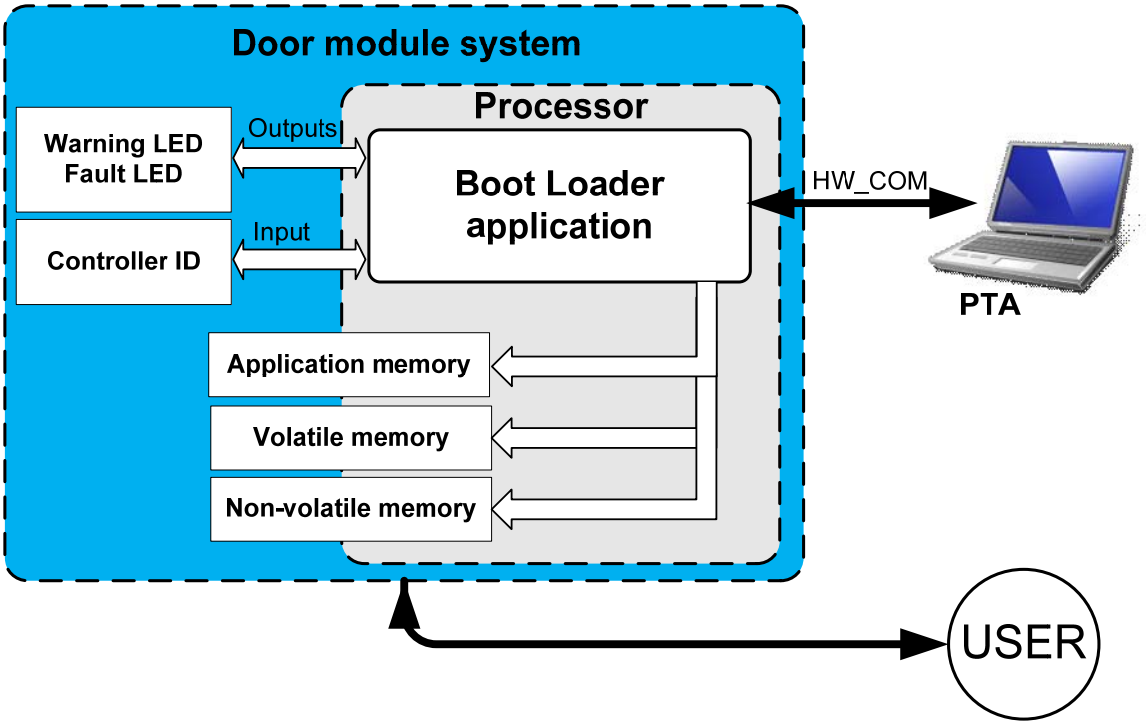

Figure 2-1: System interface diagram

2.1.2 User Interfaces

The DMS Boot Loader application user interface is limited to two LEDs:

- The Warning LED (yellow) is turned on during Boot Loader startup and self-test, and then flashes to indicate that the Boot Loader is running and waiting for PTA connection.
- The Fault LED (red) flashes following a pattern if an error is detected during self-test; otherwise it stays OFF.

2.1.3 Hardware Interfaces

The DMS Boot Loader application interfaces with the following peripherals:

- Fault LED
- Warning LED
- Hardware communication interface (HW_COM)
- Non-volatile memory
- Volatile memory

- Application memory
- Controller ID

2.1.4 Software interfaces

2.1.4.1 Applications file information

The information describing each application is contained in a structure saved in MemoryApplication, at a different address for each application. The information structure is the same for DMS and Boot Loader application, please report to Table 3-3 for a full description of the information structure.

2.1.5 Communications interfaces

The Boot Loader Application communicates with the PTA (when connected) using HW_COM. The commands applicable to this interface are:

- HW_COM.PTA.LoginRequest
- HW_COM.BootLoader.LoginResponse
- HW_COM.PTA.LogoutRequest
- HW_COM.BootLoader.LogoutResponse
- HW_COM.PTA.SoftwareVersionsRequest
- HW_COM.BootLoader.SoftwareVersionsResponse
- HW_COM.PTA.CodeStatusRequest
- HW_COM.BootLoader.CodeStatusResponse
- HW_COM.PTA.DMSApplicationUpgradeRequest
- HW_COM.BootLoader.DMSApplicationUpgradeResponse
- HW_COM.PTA.HeartbeatRequest
- HW_COM.BootLoader.HeartbeatResponse

2.1.6 Memory constraints

The Boot Loader uses less than 10% of each installed non-volatile memories.

2.1.7 Operations

Not applicable.

2.1.8 Site adaptation requirements

Not applicable.

2.2 Product functions

The following is a summary of the major functions that the software performs:

- Activate status LED.
- Provide diagnostic testing to perform low-level hardware tests.
- Provide PTA interface to upload DMS application software.

2.3 User characteristics

The intended users of the DMS should have a good understanding of how the system works, what are the modes of operation, and how to read the status of the DMS. For maintenance, the users should have technical expertise in electronics.

2.4 Constraints

The following lists the factors that limit the software developer's options:

- Software development shall conform to the software engineering process CMM-01.
- Software development shall conform to the software engineering policy manual CMM-02.
- The protocol used for the hardware interface between the PTA and Boot loader shall comply with #201111489.

2.5 Assumptions and dependencies

The following lists the factors that can affect the requirements stated in this SRS:

- Change in the features listed for the Boot Loader application of the door system.
- Change of LED configuration.
- Change of PTA protocol document 201111489.

2.6 Apportioning of requirements

All requirements are addressed in this document.

3 SPECIFIC REQUIREMENTS

3.1 External interfaces

Please refer to Figure 2-1: System interface diagram for the external interfaces of the DMS Boot Loader.

3.1.1 HW_COM

The Figure 3-1 depicts the software system logical interfaces and shows how the PTA interfaces with each Boot Loader via the HW_COM. This document covers the HW_COM communication interface only. As it may be possible for a single PTA port to communicate with more than one subsystem, the HW_COM interface protocol has the ability to specify the source/destination CPU as shown in figure 3-1. See also section 3.1.4.

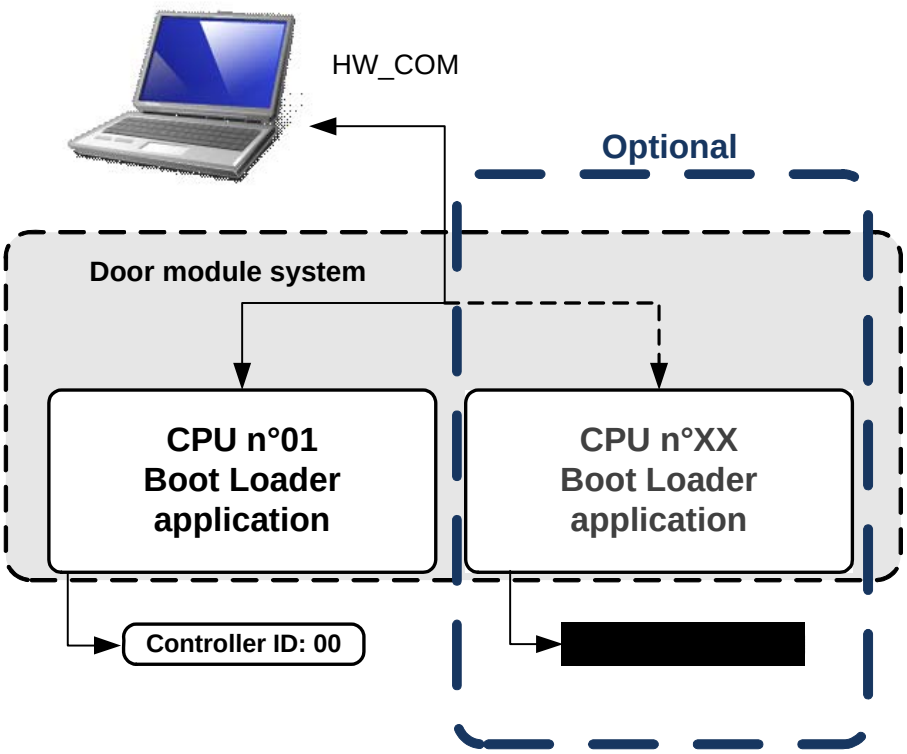

Figure 3-1: Software product perspective

Table 3-1: Mnemonics of HW_COM elements

Mnemonic	Description
HW_COM general :	
HW_COM.PTA.Message	Any single packet sent by the PTA on the HW_COM
HW_COM.BootLoader.Message	Any single packet sent by the Boot loader on the HW_COM
HW_COM Requests and Responses :	
HW_COM.PTA.LoginRequest	A request for login from the PTA to the Boot Loader on HW_COM
HW_COM.BootLoader.LoginResponse	A response to HW_COM.PTA.LoginRequest
HW_COM.PTA.LogoutRequest	A request for logout from the PTA to the Boot Loader on HW_COM
HW_COM.BootLoader.LogoutResponse	A response to HW_COM.PTA.LogoutRequest
HW_COM.PTA.SoftwareVersionsRequest	A request for DMS versions from the PTA to the Boot Loader on HW_COM
HW_COM.BootLoader.SoftwareVersionsResponse	A response to HW_COM.PTA.SoftwareVersionsRequest
HW_COM.PTA.CodeStatusRequest	A request for code status from the PTA to the Boot Loader on HW_COM
HW_COM.BootLoader.CodeStatusResponse	A response to HW_COM.PTA.CodeStatusRequest
HW_COM.PTA.DMSApplicationUpgradeRequest	A request to update the DMS application from the PTA to the Boot Loader on HW_COM
HW_COM.BootLoader.DMSApplicationUpgradeResponse	A response to HW_COM.PTA.DMSApplicationUpgradeRequest
HW_COM.PTA.HeartbeatRequest	A request of the PTA to verify the presence of the BootLoader.
HW_COM. BootLoader.HeartbeatResponse	A response to HW_COM.PTA.HeartbeatRequest

3.1.2 Memory

The PTA can interface with a maximum of 3 types of memories: Volatile Memory, Non-Volatile Memory and Application Memory. It is possible that one or more of these memories will not be used by the Boot Loader.

Table 3-2: Mnemonics of memory elements

Mnemonic	Description
Memory :	
MemoryNonVolatile	A non-volatile memory
MemoryVolatile	A volatile memory
MemoryVolatile.LoginState	Login state saved in memory
MemoryApplication	A memory containing the Boot Loader and DMS application
MemoryApplication.DMSApplication	DMS application binary saved in memory
MemoryApplication.BootLoaderApplication	Boot Loader application binary saved in memory
MemoryApplication.DMSApplication.AppInfo	DMS application information
MemoryApplication.BootLoaderApplication.AppInfo	Boot Loader application information
Memory Content :	
MemoryNonVolatile.Content	All memory cells and address lines of the MemoryNonVolatile
MemoryVolatile.Content	All memory cells and address lines of the MemoryVolatile
Memory integrity test error :	
MemoryNonVolatile.IntegrityError	Integrity check error on MemoryNonVolatile.Content
MemoryVolatile.IntegrityError	Integrity check error on MemoryVolatile.Content
MemoryApplication.DMSApplication.IntegrityError	Integrity check error on the DMS application
MemoryApplication.BootLoaderApplication.IntegrityError	Integrity check error on the Boot Loader application

Table 3-3: MemoryApplication.DMSApplication.AppInfo and MemoryApplication.BootLoaderApplication.AppInfo data format

Variable size	Variable name	Description	Offset (bytes)
unsigned long	CodeSize	Size of the SCI code	0
unsigned int	CodeCrc	CRC of the SCI code	4
unsigned char	SciIndex	Represents the SCI Index corresponding to the version number contained in the Revision[] field.	8
unsigned char	Type	This field defines the type of SCI. Possible values: 0 Firmware version, software which requires a component replacement in order to be updated (ex. boot PROM, low-level driver, etc.), 1 Software version, software which can be updated directly on the equipment using a PTE (typically the application software), 2 Externally programmable FPGA, 3 Other.	9
unsigned char	Acronym[8]	Acronym of the Item to which this configuration variable applies to	10
unsigned char	Revision[4]	version number of the SCI	18
unsigned char	Compile Date[11]	Application date of compilation	22
unsigned char	Spare	Spare byte	33
unsigned char	Compile Time[8]	Application time of compilation	34
unsigned int	uCOSII	Version of uCOSII	42
unsigned char	Reserved[4]	TDB	44

3.1.3 LED

The LEDs are the unique user interface of the Boot Loader and can be used to display information, by using different blinking pattern.

Table 3-4: Mnemonics of LED elements

Mnemonic	Description
LED :	
LEDWarning	LED used to display warning to the user
LEDFault	LED used to display faults to the user
Specific LEDFault pattern :	
LEDFault.MemoryNonVolatileIntegrityError	Specific blinking pattern of the LEDFault indicating MemoryNonVolatile.IntegrityError
LEDFault.MemoryVolatileIntegrityError	Specific blinking pattern of the LEDFault indicating MemoryVolatile.IntegrityError
LEDFault.DMSApplicationIntegrityError	Specific blinking pattern of the LEDFault indicating MemoryApplication.DMSApplicationIntegrityError

3.1.4 Inputs

In case only one CPU is present, the controller ID value will be Zero. In case more than one CPU can be addressed by a single PTA through a shared interface, the controller ID will be used to differentiate them, with a unique controller ID number for each of them.

Table 3-5: Mnemonic of inputs elements

Mnemonic	Description
Inputs	
ControllerID	Controller ID of the CPU running the Boot Loader

3.2 Functions

The following sections describe the actions that take place within the software in order to process inputs and outputs.

3.2.1 PTA Communication Functions

3.2.1.1 VAP2.36 Check incoming and outgoing message validity

a) Purpose

The purpose of this function is to check the validity of every incoming and outgoing message from and to the PTA.

b) Data

Source	Input
HW_COM	HW_COM.PTA.Message
MemoryVolatile	MemoryVolatile.LoginState
Input	ControllerID

Destination	Output
HW_COM	HW_COM.BootLoader.Message
HW_COM	Execute command from incoming message (see following functions for available commands)

c) Processing

1. VAP2.36.1 Upon reception of HW_COM.PTA.Message, the Boot Loader software shall verify the integrity of the message content, that the MemoryVolatile.LoginState value is valid (connected) and that the ControllerID of the message corresponds to the ControllerID of the CPU. If these tests are passed, the Boot Loader shall verify the validity of the command contained in the HW_COM.PTA.Message and execute the requested command.
2. VAP2.36.4 If HW_COM.PTA.Message fails one or more of the validity tests, HW_COM.PTA.Message should not be taken into account by the Boot Loader.
3. VAP2.36.5 Upon reception of an invalid command, the system shall inform the requesting end that the command is not supported.
4. VAP2.36.6 Every outgoing HW_COM.BootLoader.Message sent to the PTA by the Boot Loader shall bear the ControllerID of the CPU running the Boot Loader.

3.2.1.2 VAP2.40 Login

a) Purpose

The purpose of this function is to ensure that the PTA is logged in before answering to any other request.

b) Data

Source	Input
HW_COM	HW_COM.PTA.LoginRequest

Destination	Output
Memory	MemoryVolatile.LoginState
HW_COM	HW_COM.BootLoader.LoginResponse

c) Processing

1. VAP2.40.1 Boot Loader Application shall not accept others commands until the PTA application is logged into Door Module System.
2. VAP2.40.4 Upon reception of HW_COM.PTA.LoginRequest from the PTA, the Boot Loader shall update MemoryVolatile.LoginState (MemoryVolatile.LoginState = connected) and answer with HW_COM.BootLoader.LoginResponse
3. VAP2.40.2 Upon login, if the PTA is set in Door Module System application mode, the Boot Loader Application shall verify the Door Module System application software integrity and start the Door Module System application.

3.2.1.3 VAP2.41 Logout

a) Purpose

The purpose of this function is to logout the PTA upon request.

b) Data

Source	Input
HW_COM	HW_COM.PTA.LogoutRequest

Destination	Output
Memory	MemoryVolatile.LoginState
HW_COM	HW_COM.BootLoader.LogoutResponse

c) Processing

1. VAP2.41.1 Upon reception of the HW_COM.PTA.LogoutRequest from the PTA, the Boot Loader shall cease communication with the PTA (MemoryVolatile.LoginState = disconnected) and send HW_COM.BootLoader.LogoutResponse.
2. VAP2.41.2 Upon reception of HW_COM.PTA.LogoutRequest, the Boot Loader shall test software integrity of the Door Module System application and execute the Door Module System application if found valid. Whenever the software integrity of the current Door Module System application is not valid, the Boot Loader shall continue its execution and wait for a valid login from PTA.

3.2.1.4 VAP2.52 Automatic logout

a) Purpose

The purpose of this function is to automatically logout if no valid communication is received from the PTA for more than 1 minute.

b) Data

Source	Input
HW_COM	HW_COM.PTA.Message

Destination	Output
Memory	MemoryVolatile.LoginState

c) Processing

1. VAP2.52.1 Upon no reception of HW_COM.PTA.Message for more than 1 minute, the Boot loader shall cease communication with the PTA by logging out (MemoryVolatile.LoginState = disconnected).

3.2.1.5 VAP2.51 Verify connection state

a) Purpose
The purpose of this function is to test the state of connection state periodically.

b) Data

Source	Input
HW_COM	HW_COM.PTA.HeartbeatRequest

Destination	Output
HW_COM	HW_COM.BootLoader.HeartbeatResponse

c) Processing
1. VAP2.51.1 Every 30 seconds, the PTA shall send HW_COM.PTA.HeartbeatRequest to the Boot Loader and receive HW_COM.BootLoader.HeartbeatResponse in answer, to ensure that a working connection is still established between the Boot Loader and the PTA.

3.2.1.6 VAP2.1 Send software version

a) Purpose
The purpose of this function is to send the DMS application and Boot Loader application AppInfo upon request from the PTA.

b) Data

Source	Input
HW_COM	HW_COM.PTA.SoftwareVersionsRequest
Memory	MemoryApplication.DMSApplication.AppInfo
Memory	MemoryApplication.BootLoaderApplication.AppInfo

Destination	Output
HW_COM	HW_COM.BootLoader.SoftwareVersionsResponse

c) Processing
1. VAP2.1.3 Upon reception of HW_COM.PTA.SoftwareVersionsRequest from the PTA, the Boot Loader software shall send HW_COM.BootLoader.SoftwareVersionsResponse containing MemoryApplication.BootLoaderApplication.AppInfo and MemoryApplication.DMSApplication.AppInfo in answer.

3.2.1.7 VAP2.49 Send code status

a) Purpose
The purpose of this function is to send the code status of the DMS and Boot Loader application.

b) Data

Source	Input
HW_COM	HW_COM.PTA.CodeStatusRequest
Memory	MemoryApplication.DMSApplication.AppInfo
Memory	MemoryApplication.BootLoaderApplication.AppInfo

Destination	Output
HW_COM	HW_COM.BootLoader.CodeStatusResponse

c) Processing

1. VAP2.49.1 Upon reception of HW_COM.PTA.CodeStatusRequest, the Boot Loader shall send HW_COM.BootLoader.CodeStatusResponse containing The CRC and the size of the DMS and Boot Loader applications to the PTA Application.

3.2.1.8 VAP2.37 Program application software

a) Purpose

The purpose of this function is to upgrade the Door Module System application software.

b) Data

Source	Input
HW.COM	HW_COM.PTA.DMSApplicationUpgradeRequest

Destination	Output
HW.COM	HW_COM.BootLoader.DMSApplicationUpgradeResponse
Memory	MemoryApplication.DMSApplication

c) Processing

1. VAP2.37.1 Upon reception of HW_COM.PTA.DMSApplicationUpgradeRequest, the Boot Loader shall receive the DMS application code and test its integrity. If successful, the Boot Loader shall then program it into MemoryApplication.DMSApplication and send HW_COM.BootLoader.DMSApplicationUpgradeResponse.

3.2.1.9 VAP2.53 Program Boot Loader application

a) Purpose

The purpose of this function is to allow upgrade of the Boot Loader application.

b) Data

Source	Input
HW.COM	Boot Loader application

Destination	Output
Memory	MemoryApplication.BootLoaderApplication

c) Processing

1. VAP2.53.1 The system shall allow upgrade of the Boot Loader application through the HW_COM interface into MemoryApplication.BootLoaderApplication. The implementation of this function is specific to the CPU being used and shall be define in the design phase.

3.2.2 Self-tests functions

3.2.2.1 VAP2.44 Verify applications integrity

a) Purpose
The purpose of this function is to verify the integrity of all installed SCIs.

b) Data

Source	Input
Memory	MemoryApplication.DMSApplication
Memory	MemoryApplication.BootLoaderApplication

Destination	Output
Memory	MemoryApplication.BootLoaderApplication.IntegrityError
Memory	MemoryApplication.DMSApplication.IntegrityError

c) Processing

1. VAP2.44.1 The system shall verify the integrity of MemoryApplication.DMSApplication and MemoryApplication.BootLoaderApplication at boot-up and set the corresponding MemoryApplication.BootLoaderApplication.IntegrityError or MemoryApplication.DMSApplication.IntegrityError.
2. VAP2.44.2 The Boot Loader shall stay in boot loader mode whenever the integrity of DMS application is not valid and MemoryApplication.DMSApplication.IntegrityError shall be activated.

3.2.2.2 VAP2.50 Verify volatile memory

a) Purpose
The MemoryVolatile may be used by the Boot Loader and Door Module System Application. It must be tested before it is used.

b) Data

Source	Input
MemoryVolatile	MemoryVolatile.Content

Destination	Output
Memory	MemoryVolatile.IntegrityError

c) Processing

1. VAP2.50.1 The Boot Loader shall verify the integrity of MemoryVolatile.Content before it can use MemoryVolatile
2. VAP2.50.3 If an error occurs during MemoryVolatile.Content test, MemoryVolatile.IntegrityError shall be activated.

3.2.2.3 VAP2.45 Verify non-volatile memory

a) Purpose
The MemoryNonVolatile may be used by the Boot Loader and Door Module System Application. It must be tested before it is used.

b) Data

Source	Input
MemoryNonVolatile	MemoryNonVolatile.Content

Destination	Output
Memory	MemoryNonVolatile.IntegrityError

c) Processing

1. VAP2.45.1 The Boot Loader shall verify the integrity of MemoryNonVolatile.Content before it can use MemoryNonVolatile
2. VAP2.45.3 If an error occurs during MemoryNonVolatile.Content test, MemoryNonVolatile.IntegrityError shall be activated.
3. VAP2.45.4 MemoryNonVolatile.Content shall be preserved from corruption during integrity test

3.2.3 User interface functions

3.2.3.1 VAP2.43 Control LED

a) Purpose

The Warning LED blinks when the Boot Loader is running. The Fault LED blinks when the Boot Loader has detected a fatal error on CPU environment.

b) Data

Source	Input
Memory	MemoryNonVolatile.IntegrityError
Memory	MemoryVolatile.IntegrityError
Memory	MemoryApplication.DMSApplication.IntegrityError

Destination	Output
LED	LEDWarning
LED	LEDFault
LED	LEDFault.MemoryNonVolatileIntegrityError
LED	LEDFault.MemoryVolatileIntegrityError
LED	LEDFault.DMSApplicationIntegrityError

c) Processing

1. VAP2.43.1 The system shall blink the LEDWarning while the system is in the Boot Loader. The retained blinking pattern should be unique.
2. VAP2.43.2 If MemoryNonVolatile.IntegrityError is activated, the system shall blink the LEDFault using LEDFault.MemoryNonVolatileIntegrityError unique pattern
3. VAP2.43.3 If MemoryVolatile.IntegrityError is activated, the system shall blink the LEDFault using LEDFault.MemoryVolatileIntegrityError unique pattern.
4. VAP2.43.5 For each error to be displayed by the LEDFault, the system should use an unique pattern identifying it unambiguously .
5. VAP2.43.6 If MemoryApplication.DMSApplication.IntegrityError is activated, the system shall blink the LEDFault using LEDFault.DMSApplicationIntegrityError unique pattern .
6. VAP2.43.7 The system shall blink the LEDFault in a unique pattern if MemoryApplication.BootLoaderApplication.IntegrityError is activated.

3.3 VAP2.28 Performance requirements

This section identifies the static and dynamic numerical requirements for the Boot Loader software.

3.3.1 VAP2.28.1 Static numerical requirements

1. VAP2.28.1.4 Boot Loader shall use less than 10% of each installed memory.
2. VAP2.28.1.5 Boot Loader shall use less than 50% of available processing time.

3.3.2 Dynamic numerical requirements

No dynamic numerical requirement is used in Boot Loader

3.4 Logical database requirements

Not Applicable.

3.5 Design constraints

3.5.1 Standards compliance

- The language, and its implementation for the selected microprocessor system, shall be commercially available in English.
- All languages and operation systems must have an acceptable customer base (widespread use) and shall be commercially available

3.6 VAP2.29 Software system attributes

3.6.1 Reliability

- To ascertain that the software development is performed in a trustable technique, the development activities shall be compliant to the VSRS Software Engineering Process (SEP) CMM-01.

3.6.2 VAP2.29.1.2 Availability

- VAP2.29.1.2.1 The Boot Loader shall be fully functional in less than 60 seconds after a power-up or a reset .
- VAP2.29.1.2.2 To ensure availability, system must run under 1-second watchdog supervision; a failure to reset the watchdog timer shall trigger the re-initialization.

3.6.3 VAP2.29.1 Security

- VAP2.29.1.1 The system shall not allow access to any door control in the Boot Loader.

3.6.4 VAP2.29.2 Maintainability

- VAP2.29.2.1 The Boot Loader shall be upgraded only if a jumper is installed on a special connector on Door Module System electronic card while the Door Module System reboot.

- To ease the maintainability, the development shall be compliant to the VSRS Software Engineering Process (SEP) CMM-01 and the complete Boot Loader documentation shall be generated, inspected and released.

3.6.5 Portability

Not applicable.

3.7 VAP2.30 Other requirements

No other Requirements.